

EX NO : 5

DATE : 08-07-26

DECISION TREE MODELING

AIM:

To perform Exploratory Data Analysis (EDA) and preprocess the Titanic dataset, build a Decision Tree Classification model to predict passenger survival, visualize the decision tree, and predict the survival status for new passenger data.

PROBLEM STATEMENT:

Load the Titanic dataset from the Seaborn library. Perform Exploratory Data Analysis (EDA) to understand the dataset, identify the target variable, and preprocess the data by handling missing values and encoding categorical features. Build a Decision Tree Classifier to predict passenger survival, visualize the generated decision tree, and use the trained model to predict the survival of a new passenger.

TASKS:

1. Load the Titanic dataset from the Seaborn library.
2. Perform Exploratory Data Analysis (EDA) and display the number of rows and columns.
3. Identify the target column.
4. Check for missing values and perform the required data cleaning and preprocessing.
5. Encode categorical features into numerical values.
6. Split the dataset into training and testing sets.
7. Build a Decision Tree Classification model to predict survival.
8. Evaluate the model using accuracy.
9. Display the Decision Tree.
10. Predict the survival status for a new passenger.

CODE:

```
import pandas as pd
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.metrics import accuracy_score
import matplotlib.pyplot as plt

# Load Titanic dataset
df = sns.load_dataset('titanic')

# EDA
print(df.head())
print("\nRows and Columns:", df.shape)
print("\nTarget Column: survived")
print(df.info())
print(df.isnull().sum())

# Select required columns
df = df[['sex', 'pclass', 'age', 'sibsp', 'parch', 'fare', 'embarked', 'survived']]

# Data preprocessing
df = df.dropna()

le = LabelEncoder()
```

```

df['sex'] = le.fit_transform(df['sex'])
df['embarked'] = le.fit_transform(df['embarked'])

# Features and Target
X = df.drop('survived', axis=1)
y = df['survived']

# Split data
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Train model
model = DecisionTreeClassifier(
    criterion='entropy',
    max_depth=4,
    random_state=42
)

model.fit(X_train, y_train)

# Prediction
y_pred = model.predict(X_test)

# Accuracy
print("\nAccuracy:", accuracy_score(y_test, y_pred))

# Display Tree
plt.figure(figsize=(18,10))

plot_tree(
    model,
    feature_names=X.columns,
    class_names=['Died', 'Survived'],
    filled=True,
    rounded=True,
    fontsize=9
)

plt.show()

# Predict for new data
new_data = pd.DataFrame({
    'sex': [1],
    'pclass': [3],
    'age': [25],
    'sibsp': [0],
    'parch': [0],
    'fare': [15.5],
    'embarked': [2]
})

prediction = model.predict(new_data)

if prediction[0] == 1:
    print("\nPrediction: Survived")
else:
    print("\nPrediction: Died")

```


Figure 1


$$(x, y) = (0.9220, 0.888)$$

```
[5 rows x 15 columns]
```

```
Rows and Columns: (891, 15)
```

Target Column: survived

```
<class 'pandas.DataFrame'>
```

```
RangeIndex: 891 entries, 0 to 890
```

Data columns (total 15 columns):

#	Column	Non-Null Count	Dtype
0	survived	891 non-null	int64
1	pclass	891 non-null	int64
2	sex	891 non-null	str
3	age	714 non-null	float64
4	sibsp	891 non-null	int64
5	parch	891 non-null	int64
6	fare	891 non-null	float64
7	embarked	889 non-null	str
8	class	891 non-null	category
9	who	891 non-null	str
10	adult_male	891 non-null	bool
11	deck	203 non-null	category
12	embark_town	889 non-null	str
13	alive	891 non-null	str
14	alone	891 non-null	bool

```
dtypes: bool(2), category(2), float64(2), int64(4), str(5)
```

```
memory usage: 80.7 KB
```

```
None
survived      0
pclass        0
sex           0
age           177
sibsp         0
parch         0
fare          0
embarked      2
class         0
who           0
adult_male    0
deck          688
embark_town   2
alive         0
alone         0
dtype: int64
```

Accuracy: 0.7132867132867133

Prediction: Died